

Autodesk® Scaleform®

AMP

Scaleform AMP(Analyzer for Memory and Performance)

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: 3.02
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Autodesk® Scaleform® 4.2

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AMP

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1 소개

AMP™ Scaleform® , Scaleform CPU , , , AMP ActionScript

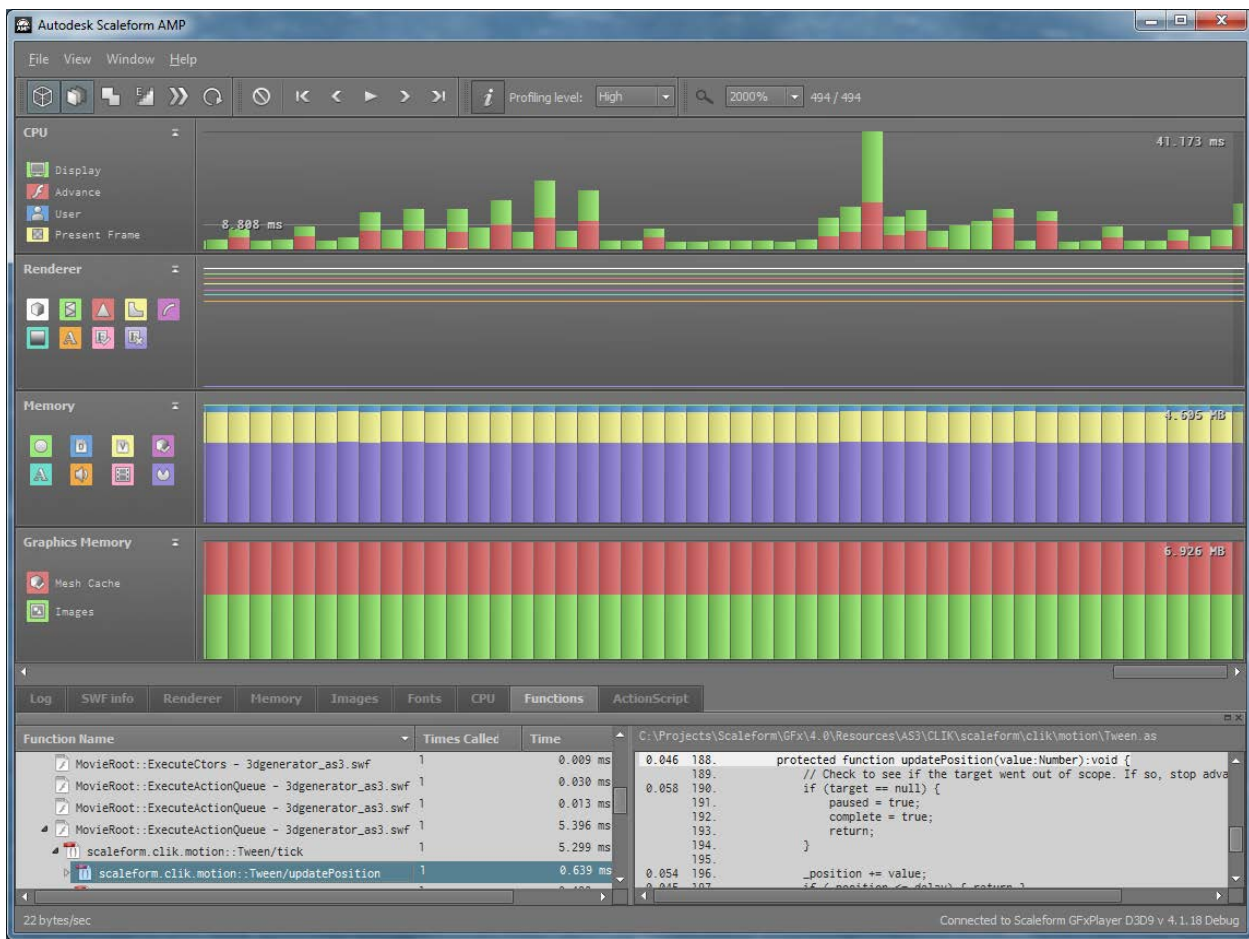


그림 1: Scaleform AMP

Scaleform AMP , AMP AMP

AMP Scaleform 3.2

1.1 AMP

Scaleform (AMP)

Scaleform . ActionScript C++

Scaleform . ActionScript 2

. (ActionScript 3

.) , flash ,

ActionScript .

APM AMP

Shipping AMP ,

AMP .

Movie . SWF

view . , AMP

Movies ActionScript .

, AMP

flash movie , , ,

(legend)

(toggle)

- **Rendering graph:** [Best Practices Guide](#)

(primitive),

)

Scaleform

- **Memory graph:** Scaleform

$$\left(\begin{array}{cccc} & & & \\ & & & \\ & & & \\ & & & \end{array} \right)$$

- **Graphics memory graph:** Scaleform

AMP

GPU

- **Log tab:** Log Scaleform

- **SWF info tab:**

Frame rate

- **Renderer tab:**

- **Memory tab:** Scaleform, Scaleform

- **Images tab:**

- **Fonts tab:**
 - Font
 - Text
 - Text (DXT)
 - Text (AMP)
 - SWF
 - Images
- **CPU tab:** Advance, Display, User
- **Functions tab:**
 - Functions
 - ActionScript C++
 - ActionScript VM
- **ActionScript tab:**
 - Functions
 - C++
- **Source pane:**
 - Functions
 - ActionScript
 - AS3
 - AS
 - SWF
 - AS3
 - AS2
 - ()
 - AS2
 - Scaleform
 - 1
 - AMP
 - (SWF)
 - flash

- SWF
 - AMP
 - AS3
- **File Menu:**
 - o **Connection:**
 - o **Disconnect:**
 - o **Debug Info Paths:**
 - AMP SWD (AS2) AS
 - (AS3) Add
 - Remove
 - o **Load Profile Frames:**
 - o **Save Profile Frames:**
 - "Load Profile Frames"
 - o **Dump Memory Report:**
 - AMP
 - "AmpMemReport.txt"
 - o **Exit:**
- **View menu:**
 - o **Graph Tooltips:**
 - AMP
 - o **Time Units:** Display Advance
 - (millisecond) (microsecond)
 - o **Memory Units:** Memory Memory
 - (byte), (kilobyte), (megabyte)

- **Window menu:** (Log, SWF Info, Renderer, Memory, Images, Fonts, CPU, Functions, ActionScript) . AMP

Restore UI .

- Help menu: . Scaleform.com

User Guide AMP

About

- **Application Control Toolbar:**

UI .

- o  **Wireframe:**

- o  **Overdraw mode:**

. AMP

(,), ,

- o  **Batch profiling mode:**

- o  **Anti-aliasing:** ()

. Scaleform EdgeAA() ,

HW FSAA() None()

- o  **Fast-forward:**

flash

. SWF

FPS

- o  **Restart:** flash

- **Frame control toolbar:**

- o  **Clear:**

AMP

- o  **First Frame:**

- o  **Previous Frame:**

- o  **Pause:**

- o  **Next Frame:**

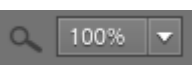
- o  **Last Frame:**

- **Profiling toolbar:**

- o  **Detailed Profiling(** **):** AMP

Memory

AMP Images 10

-  **Zoom toolbar:**

- **Status bar:**

Status Bar

AMP

UI

AMP

Window

Reset UI

Scaleform
(NDEV)
USB (COM) PC . IP "wii"
Wii AMP .

CPU CPU Summary, Functions ActionScript
 , Function
 , ActionScript

그림 4 : CPU Summary

CPU CPU ,

2.3

f

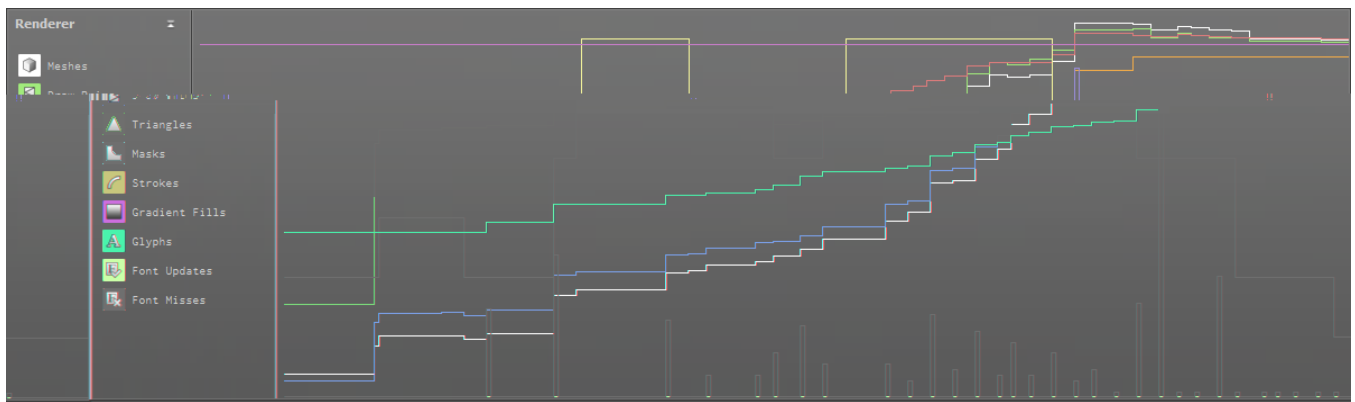


그림 5: 렌더링 그래프

Render Summary

Log	SWF info	Renderer	Memory
Meshe		182	
Draw Primitives		183	
Triangles		5124	
Masks		6	
Filters		0	
Strokes		21	
Gradient Fills		0	
Rasterized Glyphs		412	
Font Cache Updates		0	
Font Textures		1	
Font Cache Thrashing		0	
Font Cache Fill		28%	
Font Cache		1048576 pixels	
Font Cache Used		298294 pixels	
Font Failures		0	
Font Cache Misses		5	
Mesh Thrashing		0	

그림 6: Rendering Summary

2.4

Scaleform

- / /

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/

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•

“i”

.

()

- .

-

- /

- **Primitive:**
- **Fill:**
- **Mesh:**
- **MeshBatch:** (chunk)
- **Context:**
 - (thread)
- **NodeData:**
- **TreeCache:**
- **TextureManager:**
- **MatrixPool:**
- **MatrixPoolHandles:**
- **Text:** , html (line)
- **Font:**
- **MovieDef:**
 - **CharDefs:**
 - **ShapeData:**
 - **Tags:** ActionScript
 - **Fonts:**

- **Sounds:**
- **ASBinaryData:** ActionScript
- **MD_Other:** MovieDef
- o **MovieView:** ActionScript
 - **MovieClip:**
 - **ActionScript:** ActionScript
 - **ASString:** ActionScript
 - **Text:**
 - **XML:** XML
 - **MV_Other:** Movie View
 - **VM:** ActionScript
 - **AS3 VM:** ActionScript 3
 - **CallFrame:** ActionScript
 - **VTable:** ActionScript
 - **SlotInfo:** ActionScript
 - ClassTraits InstanceTraits
 - SlotInfo ClassTraits
 - InstanceTraits
 - o **SlotInfoHash:** ActionScript (hash)
 - **ClassTraits:** ActionScript
 - **Class:** ActionScript

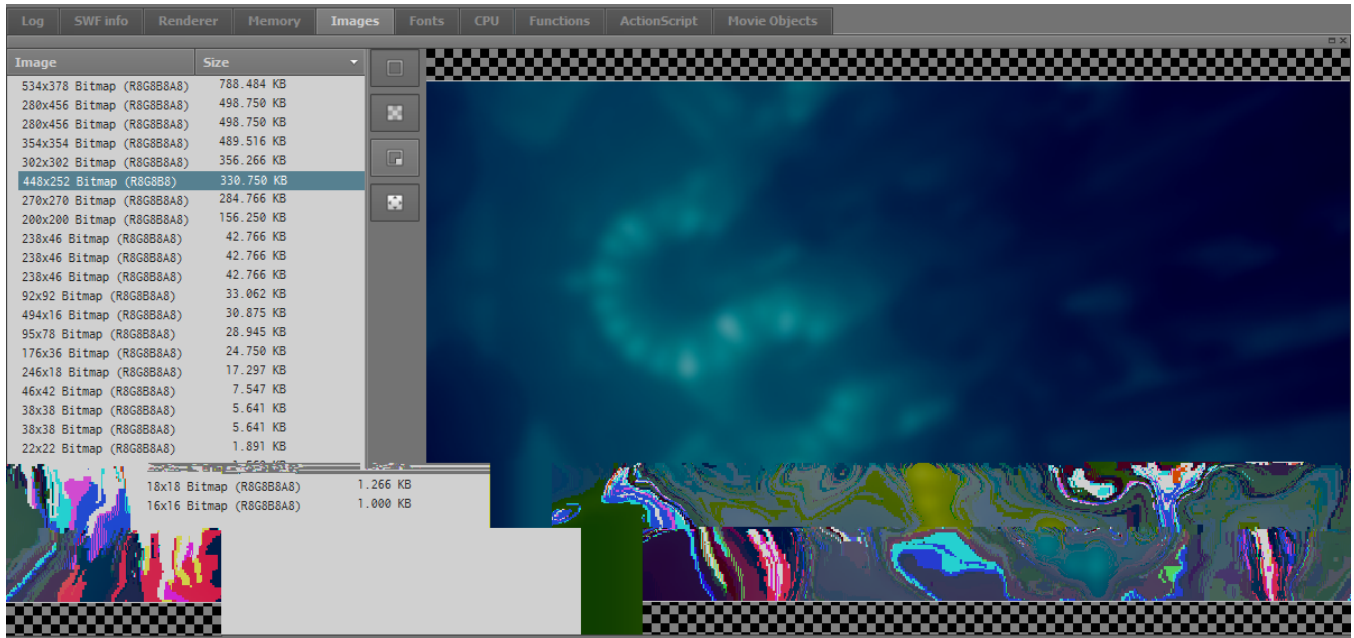
- **InstanceTraits:** ActionScript
- **Instance:** ActionScript
- **AbcFile:**
 - **AbcConstPool:** , , Namespaces Multinames
(constants)
AbcFile
 - **VMAbcFile:** AbcFiles ,
 - **Tracer:**
 - o **IME:** Input Method Editor (IME)

Log	SWF info	Renderer	Memory	Image
Memory		Size		
▲	Total Footprint	4849.402 KB		
▲	Used Space	4633.715 KB		
▷	Video Heaps	0.000 KB		
▷	Other Heaps	325.621 KB		
▲	Movie View Heaps	1226.094 KB		
▲	MovieView "3dgenerator_as3.swf"	1226.094 KB		
	String	0.828 KB		
▷	Renderer	12.672 KB		
▷	MovieView	1176.781 KB		
	General	35.812 KB		
▷	Movie Data Heaps	261.066 KB		
▲	Global Heap	2820.934 KB		
	Video	264.176 KB		
	String	20.633 KB		
	Sound	0.141 KB		
▷	Renderer	2310.914 KB		
▷	MovieView	8.039 KB		
▷	MovieDef	116.676 KB		
	Image	0.141 KB		
	IME	9.914 KB		
	General	90.301 KB		
▷	Debug Memory	8.000 KB		
▷	Unused Space	38.820 KB		
	Debug Data	176.867 KB		

그림 8: Memory Summary

Scaleform

Images



크기 9: Images 메모리 탭

Scaleform

..

Scaleform

Fonts



그림 10: Fonts 메모리 탭

SWF

, MovieData

CharDefs

AMP

Heaps

Memory Summary

SWF

2.5

AMP

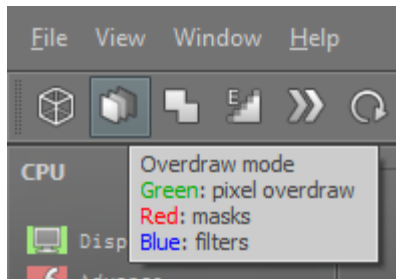


그림 11: 전환 버튼

(
16), , (8) . ,

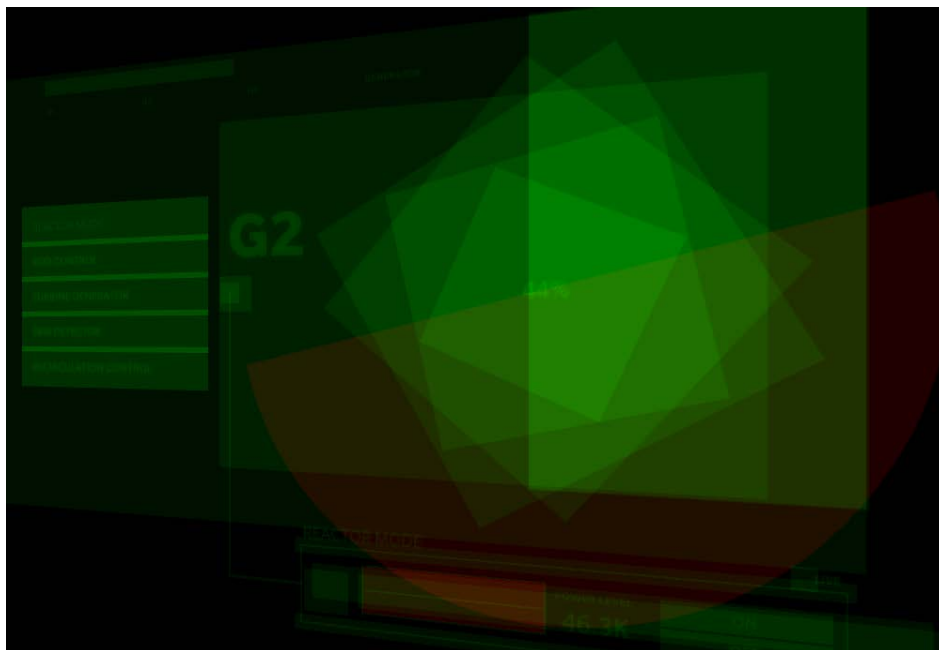


그림 12: Highlighting 을 위한 렌더링 모드

2.6

AMP

Figure 13: Batch profiling mode

3 플랫폼 및 통합 관련 주의사항

- AMP
 - GFx::System GFx::System::Init
 - Scaleform::System Scaleform
 - GFx::AMP::Server::Init() GFx::AMP::Server::Uninit()
 - AMP
- AMP
 - Windows, Xbox360, PS3, Linux, Mac, Wii, Android iPhone/iPad, 3DS, PS Vita, WiiU
 - AMP
 - AMP IP , IP
 - Xbox360 IP IP . IP
 - Xbox 360 Neighborhood
 - PS3 IP IP
 - IP
 - Android
 - AndroidManifest.xml
 - ```
<uses-permission android:name="android.permission.INTERNET" />
```
      - WiFi PC WiiU . PC
        - 6003 . TCP/IP
      - Wii AMP . NDEV
        - USB (COM ) PC . IP
        - "wii" Wii AMP
        - : RVL\_SDK/X86/bin . hio2.dll
        - AMP
  - Unreal Engine 3
    - AMP Flash
    - SWD/AS3 AS . AMP SWF

AS2 SWD

SWD/AS

AMP

.

## 4 빌드 주의사항

- AMP non-shipping Scaleform .  
Scaleform AMP GfxConfig.h SF\_AMP\_SERVER Gfx  
. Scaleform  
AMP . Ptr<Gfx::AMP::ServerState> serverState = \*SF\_NEW  
Gfx::AMP::ServerState();  
serverState->StateFlags &= Gfx::AMP::Amp\_Disabled;  
AmpServer::GetInstance().UpdateState(serverState);
- AMP .  
(Windows: Ws2\_32.lib,  
Xbox 360: Xnet.lib, PS3: libnetctl, Linux: ).
- AMP , AMP  
.  
.  
.  
AMP  
AMP  
. AMP::Server::GetInstance().SetInitSocketLib(false).  
AMP .
- AMP . Scaleform  
.  
AMP::Server::GetInstance().SetHeapLimit() .
- AMP 7534 .  
. AMP::Server::GetInstance().SetListeningPort();

## 6. AMP

```
. AMP::Server::GetInstance().SetConnectionWaitTime.
```

## 5 AMP 지원 추가

Scaleform  
AMP  
Scaleform Player

### 5.1 Debug

AMP ActionScript 2 Flash (SWD )  
Flash CS3 CS4 (Ctrl+Shift+Enter) SWD  
SWD SWF AMP

ActionScript 3 SWD SWF  
AS Flash Studio SWF

### 5.2

AMP  
AMP

AMP

- AMP::AppControlInterface  
AMP AMP::MessageAppControl HandleAmpRequest

- AMP::Server::GetInstance().SetAppControlCallback  
.
- AMP::Server::GetInstance().SetAppControlCaps AMP  
.  
AMP::MessageAppControl , true  
:  
  
AMP::MessageAppControl caps;  
caps.SetCurveToleranceDown(true);  
caps.SetCurveToleranceUp(true);  
caps.SetNextFont(true);  
caps.SetRestartMovie(true);  
caps.SetToggleAaMode(true);  
caps.SetToggleAmpRecording(true);  
caps.SetToggleFastForward(true);  
caps.SetToggleInstructionProfile(true);  
caps.SetToggleOverdraw(true);  
caps.SetToggleBatch(true);  
caps.SetToggleStrokeType(true);  
caps.SetTogglePause(true);  
caps.SetToggleWireframe(true);  
AMP::Server::GetInstance().SetAppControlCaps(&caps);

### 5.3

AMP

AMP::Server::GetInstance().SetConnectedApp

AMP

- ServerStateType  
AMP::Server::GetInstance().SetState.  
o Amp\_RenderOverdraw

- o Amp\_App\_Wireframe
- o Amp\_App\_Paused
- o Amp\_App\_FastForward
- AMP::Server::GetInstance().SetAaMode
- AMP::Server::GetInstance().SetStrokeType
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::ServerState
- AMP::Server::GetInstance().UpdateState

## 5.4

AMP C++ ActionScript CPU  
 . ActionScript

Amp.addMarker(1)

: Amp.addMarker() .as

Resources/AS2/CLIK

Amp

C++

Gfx::MovieImpl:: AdvanceStats->AddMarker(1)

AMP

## 6 AMP 관련 FAQ

AMP

### 1. 게임 측에서 AMP 를 사용하는 데 필요한 최저 사양은 무엇입니까?

GFX\_AMP\_SERVER

AMP

GfX

AMP

. Platform::RenderThread::drawFrame

AmpServer::GetInstance().AdvanceFrame

AMP

### 2. AmpClient 를 사용하여 게임을 연결하면 게임 프레임률이 현저하게 떨어집니다. 조사해보니 AmpServer::AdvanceFrame()을 실행하면 시간이 오래 걸리는데 이게 정상입니까?

AMP

AMP

"i"( I)

. AMP

### 3. Windows/PS3/Xbox 에서 게임을 실행합니다. 검색창에서 AMP 클라이언트를 검색해 AMP 에 연결할 수는 있습니다. 그런데 통계 정보가 전혀 없습니다.

AMP

Platform::RenderThread::drawFrame

AmpServer::GetInstance().AdvanceFrame

AMP

### 4. (AMP::Server::WaitForAmpConnection 을) 업데이트할 때 왜 AMP 가 일 초 이상 멈춥니까?

AMP

AMP

Scaleform

AMP

'i'

. AMP

'profile level'

1MB

GFx::AMP::Server::SetHeapLimit

##### 5. 프로파일러를 사용할 때 예상되는 메모리 오버헤드는 어느 정도입니까?

AMP

1MB

AmpServer::SetHeapLimit

AMP

GFx

## 7 추가 정보

AMP

- [AMP Use Cases](#) .
- [Scaleform Reference](#) – – /
- [AMP Forum](#) –